

Kubernetes

Deployment & DevOps Notes — Knowledge Base

1. What is Kubernetes?

Kubernetes (K8s) is an open-source container orchestration platform, originally developed by Google, used to automate the deployment, scaling, and management of containerized applications across clusters of machines.

2. Core Objects/Concepts

- Pod — the smallest deployable unit; wraps one or more tightly coupled containers.
- Node — a physical or virtual machine that runs pods (worker nodes managed by a control plane).
- Deployment — declaratively manages a set of replica pods, handling rolling updates and rollbacks.
- Service — a stable network endpoint that load-balances traffic across a set of pods.
- ConfigMap/Secret — externalized configuration and sensitive data, kept separate from application images.
- Namespace — a way to logically partition a cluster's resources (e.g., dev, staging, prod).

3. Control Plane vs Worker Nodes

Component	Role
API Server	Front-end to the cluster; all interactions go through it
etcd	Distributed key-value store holding cluster state
Scheduler	Assigns pods to nodes based on resources/constraints
Controller Manager	Runs controllers that reconcile actual state to desired state
kubelet (on nodes)	Ensures containers are running as expected on that node

4. Example Deployment YAML

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: web-app
spec:
  replicas: 3
  selector:
    matchLabels:
      app: web
  template:
```

```
metadata:
  labels:
    app: web
spec:
  containers:
  - name: web
    image: myapp:1.0
    ports:
    - containerPort: 3000
```

5. Why Kubernetes?

- Self-healing — automatically restarts failed containers and reschedules pods from failed nodes.
- Horizontal auto-scaling based on CPU/memory or custom metrics.
- Declarative configuration — describe the desired state; Kubernetes continuously reconciles reality to match it.
- Rolling updates and easy rollbacks minimize downtime during deployments.
- Rich ecosystem: Helm (package manager), Istio (service mesh), Prometheus (monitoring).

6. Considerations

- Significant learning curve and operational complexity — often overkill for small applications.
- Managed offerings (EKS, GKE, AKS) reduce operational burden versus self-hosting a cluster.